

Energy Ledgers for Forced Harmonic ODEs

Kirchhoff power balance and an unknown component with $X = LRC$

Hob Nilre & Bo C. Herlin

2026-10-02

Abstract

An added term Xq''' supplies or absorbs energy according to the sign of $Xq'''q'$. Exact trajectories show both roles. Choosing $X = LRC$ scales the transfer on fixed motion; it does not choose its direction. Brief hammer and rotary examples apply the same rule. The component's physical identity and remaining energy account stay open.

PDF created: 2026-10-08 13:12:50 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-ode-energy>

1 From voltage balance to power accounts

Start with the forced [LRC template](#), Section 2:

$$Lq'' + Rq' + \frac{q}{C} = f(t), \quad L, C > 0, \quad R \geq 0. \quad (1)$$

Here q is charge, $i = q'$ current, f forcing voltage, and L, R, C inductance, resistance and capacitance. Coefficients are constant, trajectories smooth, and primes denote time derivatives. In the [series convention](#), Section 1, the same current multiplies every voltage term. Put the source first and record the circuit power balance in line 2, with opposite transfers in lines 1 and 3:

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0. \end{cases} \quad (2)$$

Line 1 belongs to the heat recipient and line 3 to the source or sink; $fi < 0$ means the source receives energy. Opposite entries record each transfer on both sides. Constant total energy also requires the omitted energy rates and reservoir laws; prescribed $f(t)$ alone gives no source capacity.

2 When the added term supplies or absorbs

Add a fictional component with constant $X > 0$ and unknown physical identity:

$$Xq''' + Lq'' + Rq' + \frac{q}{C} = f(t). \quad (3)$$

Treating Xq''' as its voltage contribution adds a fourth account:

$$\begin{cases} 1: & \dots + Rq'i + \dots = 0, \\ 2: & f(t)i - Xq'''i - Lq''i - Rq'i - qi/C = 0, \\ 3: & -f(t)i + \dots = 0, \\ 4: & \dots + Xq'''i + \dots = 0. \end{cases} \quad (4)$$

Thus the received power and the contribution to line 2 are

$$P_X = Xq'''i, \quad P_{X \rightarrow 2} = -Xq'''i. \quad (5)$$

Since $X > 0$, the component supplies power when $q'''i < 0$, absorbs it when $q'''i > 0$, and transfers none when the product is zero. Over an interval,

$$W_X(t_0, t_1) = \int_{t_0}^{t_1} Xq'''(t)i(t) dt, \quad W_{X \rightarrow 2} = -W_X. \quad (6)$$

Positive W_X is net received energy; negative W_X is net supplied energy. The integral fixes the transfer, while the rest of line 4 remains open.

Take $L = 1 \text{ H}$, $R = 1 \text{ } \Omega$, $C = 1 \text{ F}$ and $X = 1 \text{ H s}$. Put $u = t/(1 \text{ s})$ and obtain $f(t)$ by substituting each chosen trajectory into (3).

Supplier. For

$$q = (1 \text{ C}) \sin u, \quad i = (1 \text{ A}) \cos u, \quad q''' = -(1 \text{ A/s}^2) \cos u, \quad (7)$$

the inductive and capacitive terms cancel, as do the resistive and X terms, giving $f(t) = 0$. Hence

$$P_X = -(1 \text{ W}) \cos^2 u, \quad W_X(0, 2\pi \text{ s}) = -(1 \text{ J}) \int_0^{2\pi} \cos^2 u du = -\pi \text{ J}. \quad (8)$$

At $t = 0$, the component supplies 1 W ; over a cycle it supplies $\pi \text{ J}$. Here $P_X = -Ri^2$ at every instant, so this transfer covers the resistor's absorption. The equation leaves its physical source unspecified.

Absorber. On $0 \leq t \leq 1 \text{ s}$, choose

$$q = (1 \text{ C}) \frac{u^3}{6}, \quad i = (1 \text{ A}) \frac{u^2}{2}, \quad q''' = 1 \text{ A/s}^2. \quad (9)$$

Now

$$f(t) = (1 \text{ V}) \left(1 + u + \frac{u^2}{2} + \frac{u^3}{6} \right), \quad P_X = (1 \text{ W}) \frac{u^2}{2}, \quad (10)$$

and

$$W_X = (1 \text{ J}) \int_0^1 \frac{u^2}{2} du = \frac{1}{6} \text{ J}. \quad (11)$$

At $t = 1 \text{ s}$, $P_{X \rightarrow 2} = -1/2 \text{ W}$. The same positive coefficient supplies energy on the sinusoid and absorbs it on this polynomial.

3 What choosing $X = LRC$ changes

For $(a, b, c) = (L, R, 1/C)$, *ODE Coefficient Synthesis*, Sections 1–3, selects

$$X = A_3 = A_{-1,3} = \frac{ab}{c} = LRC > 0. \quad (12)$$

Here $R > 0$. Since RC has units of time, $[X] = \text{H s}$ and Xq''' is a voltage. Equation (5) becomes $P_{X \rightarrow 2} = -LRC q''' i$.

Fix $q(t)$, including amplitude and time scale. Increasing LRC then strengthens whichever direction $q''' i$ already selects. The signed work scales the same way on a fixed interval. With independent positive bounds, the greatest magnitude uses all three upper bounds; without bounds there is no finite maximum. Recompute $f(t)$ to retain the chosen motion.

For $q = Q \sin(\omega t)$, with $Q, \omega > 0$,

$$q''' = -\omega^2 i, \quad P_{X \rightarrow 2} = LRC \omega^2 i^2 \geq 0. \quad (13)$$

All positive choices make X a supplier except at current zeros. Absorption needs a different derivative sign, as in (9). Relative to resistor absorption, at $i \neq 0$,

$$\frac{|P_{X \rightarrow 2}|}{Ri^2} = LC \left| \frac{q'''}{i} \right|. \quad (14)$$

This ratio is $LC\omega^2$ on a sinusoid. Increasing R scales both powers equally; only L and C change their ratio on fixed motion.

Use Section 2's supplier at $t = 0$ and absorber at $t = 1$ s, recomputing the forcing for each row.

Table 1. Signed contributions $P_{X \rightarrow 2}$ on the same two trajectories.

(L, R, C) in (H, Ω , F)	X (H s)	Supplier (W)	Absorber (W)
(1, 1, 1)	1	+1	−1/2
(1, 2, 1)	2	+2	−1
(2, 2, 2)	8	+8	−4

With $1 \leq L/(1 \text{ H}), R/(1 \Omega), C/(1 \text{ F}) \leq 2$, the last row maximizes both magnitudes. The same integrals give 8π J supplied per sinusoidal cycle and $4/3$ J absorbed over the polynomial interval.

If forcing and initial conditions are fixed instead, changing the coefficients changes the motion too. Maximizing LRC alone then says nothing about the maximum transfer.

4 Hammer and nail

Let x be hammer-face and nail-head movement during smooth contact, $v = x'$, and $m, d, k > 0$ mass, effective resistance and stiffness. The *domain map*, Sections 2–3, replaces (L, R, C) by $(m, d, 1/k)$:

$$X_h x''' + mx'' + dx' + kx = F(t), \quad X_h = \frac{md}{k}. \quad (15)$$

Assume the unknown component acts inside the hammer. Its received power is $X_h x''' v$; Section 3's fixed-motion scaling becomes md/k .

For $F = 0$, $x(0) = 0$, $v(0) = v_0 > 0$ and $x''(0) = 0$, put $\omega = \sqrt{k/m}$. Up to the first stop,

$$x(t) = \frac{v_0}{\omega} \sin(\omega t), \quad 0 \leq t \leq t_* := \frac{\pi}{2\omega}. \quad (16)$$

Indeed, $mx'' + kx = 0$ and $X_h x''' + dv = 0$, so

$$W_{X_h} = \int_0^{t_*} X_h x''' v dt = - \int_0^{t_*} dv^2 dt = - \frac{\pi dv_0^2}{4\omega} = -W_d. \quad (17)$$

The component supplies the resistance work from its unspecified account. Movement x includes elastic support motion; permanent nail depth is undetermined.

5 Rotary output on prescribed motion

For output angle θ and speed $\Omega = \theta'$, substitute $(q, L, R, C, f) \mapsto (\theta, J, d, 1/k, \tau_h)$ in (3). Here $J, d, k > 0$ are output inertia, damping and torsional stiffness; τ_h is prescribed hammer torque and J excludes the hammer. The assumed component inside the tool has $X_r = Jd/k$. On identical motion, let τ_X and τ_0 be the required torques with and without it. Their input-work difference is

$$\Delta W_h = \int_{t_0}^{t_1} (\tau_X - \tau_0) \Omega dt = \int_{t_0}^{t_1} X_r \theta''' \Omega dt = W_{X_r}. \quad (18)$$

Supplied work therefore reduces the required hammer input on that motion.

For $\theta = \Theta \sin(\omega t)$, $\Theta, \omega > 0$, and $T = \pi/(2\omega)$, the relation $\theta''' = -\omega^2 \Omega$ gives

$$\begin{aligned} W_{X_r} &= \int_0^T X_r \theta''' \Omega dt = - \frac{\pi X_r \Theta^2 \omega^3}{4}, \\ W_d &= \int_0^T d \Omega^2 dt = \frac{\pi d \Theta^2 \omega}{4}, \quad \frac{-W_{X_r}}{W_d} = \frac{J \omega^2}{k}. \end{aligned} \quad (19)$$

When $J \omega^2/k \leq 1$, an ordinary model gives the same torque on this motion with nonnegative damping

$$d_{\text{eff}} = d - X_r \omega^2 = d \left(1 - \frac{J \omega^2}{k} \right). \quad (20)$$

For example, $J \omega^2/k = 1/10$ gives $d_{\text{eff}} = 9d/10$ and $W_{X_r} = -W_d/10$. The component supplies work that reduced damping avoids dissipating. This comparison adjusts the forcing to match motion; hammer preparation and the unknown account remain outside it.

References

1. H. Nilre and B. C. Herlin (2026). [The ODE Template and Its Domain Equivalents](#).
2. H. Nilre and B. C. Herlin (2026). [ODE Coefficient Synthesis](#).
3. H. Nilre and B. C. Herlin (2026). [Interconnecting Series and Parallel ODEs](#).